Kelvon Dixon

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Education

North Carolina State University — Raleigh, NC

Bachelor of Science in Computer Science, Minor in Mathematics — *Expected Graduation May 2029*

- Relevant Coursework: Discrete Math, Probability and Statistics, Software Development Fundamentals, Intro to R/Python

Montgomery County Community College — Troy, NC

Associates of Science — *May 2025*

Experience

Undergraduate Research Assistant — Jan 2026 – Present

North Carolina State University — *Raleigh, NC*

- Conduct literature review on software quality research, focusing on technical debt, code smells, and software refactoring as indicators and drivers of maintainability and long-term code health.
- Analyzed multiple open-source GitHub codebases to identify common code smells and evaluate their relationship to defect frequency, file churn, and maintainability trends over time.
- Wrote Python scripts to automate the identification of code smells and track refactoring frequency across multiple open-source repositories, reducing manual analysis time and establishing a structured data pipeline for long-term maintainability metrics.

IT Intern — May 2026 – Present

North Carolina State University — *Raleigh, NC*

- Provide comprehensive technical support for the College of Sciences, troubleshooting hardware, software, and network issues for a diverse user base of over 4,000 students, faculty, and staff.
- Streamline IT onboarding and deployment processes by configuring system images and provisioning devices, reducing standard setup times for incoming staff

Projects

Personal Library Management System | *Python, Flask, Vite, requests* — Jan 2026 – March 2026

- Developed full-stack book management application with Flask REST API and React frontend, enabling CRUD operations for 60+ book capacity with real-time statistics"
- Integrated Google Books API with secure API key authentication for automated book imports with auto-populated metadata, reducing manual entry time by 80% through environment-based credential management
- Implemented data visualization dashboard using Recharts library to display reading analytics including progress tracking and genre distribution"

Full-Stack Screen Time Quality Tracker | *Python, Flask, Javascript, SQLite, React* — Dec 2025 - Jan 2026

- Built full-stack web application using Flask and React to track and analyze screen time quality through weighted activity scoring, serving 13 RESTful API endpoints with comprehensive CRUD operations
- Engineered weighted scoring algorithm calculating daily quality scores from activity durations and quality ratings, reducing manual tracking time by 90%
- Developed interactive analytics dashboard with Chart.js visualizations (line, doughnut, bar charts) displaying weekly trends, category distributions, and session breakdowns

Technical Skills

Languages: Python, JavaScript, SQL, C/C++, HTML/CSS, R, Java
Frameworks and Libraries: React, Node.js, Express.js, Flask, Axios
Developer Tools: VS Code, Git, Github, Postman/Curl, pip, npm, Vite